Matrix Theory I: Matrices, Trace and Determinants

Problem sheet for the Day 1 class · CSIR NET / GATE / IIT JAM

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. CSIR NET June 2020 · MCQ

Let A be a 2×2 real matrix with $\det A = 1$ and $\operatorname{tr} A = 3$. The value of $\operatorname{tr}(A^2)$ is

(A) 2
(B) 10
(C) 9
(D) 7

2. CSIR NET June 2020 · MCQ

Let A and B be 2×2 matrices. Which of the following is true?

(A) $\det(A+B) + \det(A-B) = \det A + \det B$
(B) $\det(A+B) + \det(A-B) = 2\det A - 2\det B$
(C) $\det(A+B) + \det(A-B) = 2\det A + 2\det B$
(D) $\det(A+B) - \det(A-B) = 2\det A - 2\det B$

3. CSIR NET Dec 2019 · MCQ

For $t \in \mathbb{R}$ let $M(t) = \begin{pmatrix} 1 & t & 0 \\ 1 & 1 & t^2 \\ 0 & 1 & 1 \end{pmatrix}$. Which of the following is true?

(A) $\det M(t)$ is a polynomial of degree 3 in t
(B) $\det M(t) = 0$ for all t
(C) $\det M(t) = 0$ for infinitely many t
(D) $\det M(t) = 0$ for exactly two values of t

4. JAM 2019 · NAT

Let M and N be 4×4 matrices with integer entries such that $MN = 2\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. The maximum possible value of $\det M + \det N$ is _____.

5. CSIR NET · MCQ

Let A be a 5×5 real matrix such that the sum of the entries in each row of A is 1. Then the sum of all the entries of A^3 is

(A) 3
(B) 15
(C) 5
(D) 125

6. Exam pattern MSQ Let $A \in M_n(\mathbb{R})$. Which of the following are necessarily true?

(A) if n is odd and $A^T = -A$, then $\det A = 0$

(B) if $A^T A = I$ and $\det A = -1$, then $\det(I + A) = 0$

(C) if $A^k = 0$ for some k, then $\det(I + A) = 1$

(D) if $A^2 = I$, then $A = \pm I$

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7. Exam pattern NAT Let A be a 3×3 real matrix with $\det A = 2$. Then $\det\left(\operatorname{adj}(\operatorname{adj} A)\right)$ equals _____.

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8. Exam pattern NAT Let A be the 4×4 matrix with every diagonal entry 3 and every off-diagonal entry 1. Then $\det A$ equals _____.

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9. Exam pattern NAT The number of 2×2 matrices with every entry in $\{0, 1\}$ that are invertible (as real matrices) is _____.

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10. Exam pattern MSQ Let $A, B, C \in M_n(\mathbb{R})$. Which of the following are always true?

(A) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$

(B) $\det(AB) = \det(BA)$

(C) $AB - BA \neq I$

(D) $\operatorname{tr}(ABC) = \operatorname{tr}(CBA)$
